

# Elaine Jose

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## EDUCATION

### Karunya Institute of Technology and Sciences

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.22/10

Coimbatore, India

Aug. 2024 – May 2028

**Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

## EXPERIENCE

### Software Engineering Intern

May 25, 2026 – June 25, 2026

Nest Technologies

On-site Internship

- Designed and implemented an end-to-end machine learning pipeline for student placement prediction, covering data preprocessing, feature engineering, model training, evaluation, and deployment.
- Built a custom Exploratory Data Analysis (EDA) notebook from scratch to analyze feature distributions, correlations, missing values, and data quality for model development.
- Developed and evaluated Logistic Regression, Decision Tree, Random Forest, and XGBoost models, selecting Logistic Regression as the final deployment model with **79.45%** accuracy.
- Deployed the trained model using FastAPI and Joblib, integrating a responsive frontend for real-time placement prediction through REST APIs.

### Artificial Intelligence Intern

June 1, 2025 – June 30, 2025

CodTech IT Solutions (Virtual Internship)

Remote

- Developed AI-based applications including a Speech Recognition System and an automated Text Summarization Tool using Python and Natural Language Processing (NLP).
- Implemented speech-to-text and text summarization pipelines with data preprocessing and model integration.
- Built and tested machine learning applications while following Git-based version control and software development best practices.
- Improved debugging and problem-solving skills by implementing and optimizing AI-based software solutions.

## PROJECTS

### SafePath Scanner (Smart India Hackathon) | Python, FastAPI, React, Open3D, PyTorch | [Repository](#)

- Developed an AI-assisted disaster response platform for generating safe evacuation routes inside collapsed buildings using LiDAR point cloud data.
- Implemented digital twin generation, hazard detection, occupancy mapping, and A\* path planning algorithms for emergency navigation.
- Built RESTful APIs using FastAPI to enable communication between NVIDIA Jetson devices and the mobile application.
- Designed a scalable backend architecture integrating AI models, sensor fusion, and real-time visualization.

### ATRIVA – AI Critical Care Prognosis Dashboard | Python, FastAPI, React, PyTorch, WebSockets | [Repository](#)

- Developed a multimodal AI-powered healthcare dashboard for predicting Code Blue risk using ECG signals and patient vital signs.
- Built deep learning models combining CNN, BiLSTM, and GRU architectures for patient risk prediction.
- Implemented FastAPI backend services with WebSocket support for real-time telemetry streaming.
- Designed an interactive React dashboard and integrated LLM-powered clinical note generation for automated medical reporting.

### AI-Powered Placement Prediction Platform | Python, FastAPI, Scikit-Learn, Pandas, NumPy, XGBoost | [Repository](#)

- Developed an end-to-end machine learning platform for predicting student placement outcomes.
- Built a custom Exploratory Data Analysis (EDA) pipeline covering statistical analysis, feature engineering, preprocessing, correlation analysis, and visualization.

- Trained and compared Logistic Regression, Decision Tree, Random Forest, and XGBoost models, achieving **79.45%** accuracy with Logistic Regression.
- Deployed the trained model using Joblib and FastAPI, exposing REST APIs for real-time predictions through a responsive web application.

#### **AI Productivity Toolkit** | *Python, NLP, SpeechRecognition, Transformers* | [Repository](#)

- Developed an AI-powered productivity suite featuring real-time speech recognition and automatic text summarization capabilities.
- Implemented speech-to-text conversion, audio preprocessing, transformer-based summarization, and NLP pipelines for intelligent text processing.
- Designed modular Python applications following reusable software engineering principles for maintainability and scalability.
- Optimized application performance through debugging, testing, and iterative model improvements during the internship.

## TECHNICAL SKILLS

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**Programming Languages:** Python, Java, C, SQL

**Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

**Frameworks & Technologies:** FastAPI, React, REST APIs

**Machine Learning & AI:** PyTorch, Scikit-Learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Open3D

**Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook, Postman

## ACHIEVEMENTS

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- Secured **5th Place** at the **Aurelion National Hackathon 2026** among 600+ participating teams for developing the ATRIVA AI healthcare platform.
- Selected as a **Grand Finalist** in the **Smart India Hackathon (SIH) 2025**.
- Completed the **NVIDIA AI Workshop**, gaining hands-on experience with modern AI technologies and workflows.
- Completed the **Be10X AI Bootcamp**, focusing on AI tools and practical software applications.